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An empirical study on the joint impact of feature selection and data resampling on imbalance classification

Chongsheng Zhang[1] · Paolo Soda[2,3] · Jingjun Bi[1] · Gaojuan Fan[1] · George Almpanidis[1] · Salvador García[4] · Weiping Ding[5]

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Abstract
Many real-world datasets exhibit imbalanced distributions, in which the majority classes have sufficient samples, whereas the minority classes often have a very small number of samples. Data resampling has proven to be effective in alleviating such imbalanced settings, while feature selection is a commonly used technique for improving classification performance. However, the joint impact of feature selection and data resampling on two-class imbalance classification has rarely been addressed before. This work investigates the performance of two opposite imbalanced classification frameworks in which feature selection is applied before or after data resampling. We conduct a large-scale empirical study with a total of *9225* experiments on *52* publicly available datasets. The results show that both frameworks should be considered for finding the best performing imbalanced classification model. We also study the impact of classifiers, the ratio between the number of majority and minority samples (IR), and the ratio between the number of samples and features (SFR) on the performance of imbalance classification. Overall, this work provides a new reference value for researchers and practitioners in imbalance learning.

Keywords Imbalanced classification · Feature selection · Data selection · Resampling

1 Introduction

Class imbalance refers to the case where a few categories have a significantly inadequate number of samples compared to others. Imbalanced data exists in a large number of domains and applications, such as banking fraud detection, medical diagnosis, text classification, software defect prediction, financial distress prediction [26], etc. Its pervasiveness and immediate relevance to many real-world applications have attracted substantial research effort in the last two decades. There are many techniques for combating the class imbalance problem in the literature, including internal approaches that tailor an algorithm to imbalanced data, data resampling approaches [21], cost-sensitive learning, and ensemble learning.

Feature selection/dimension reduction is often an effective technique in machine learning for improving classification performance; it also simplifies the classification model to be built. However, inappropriate reduction in features may lead to a loss in the discrimination power. *Watanabe's ugly duckling theorem* [30] supports the need for a careful choice of the features, stating that it is possible to make two arbitrary patterns similar by encoding them with a sufficiently large number of redundant features.

Up to now, there are only a few preliminary studies that have explored the feature selection before data resampling (FS+DS for short) framework for improving imbalance classification performance, while the alternative feature selection after data resampling (DS+FS) framework has often been neglected. In this work, we investigate the joint impact of feature selection and data resampling for deriving the best imbalance classification model, as depicted in Fig. 1. We provide a comprehensive empirical study on *52* benchmark datasets with imbalanced distributions, using *9* feature selection methods, *6* resampling approaches for class imbalance learning, and *3* representative baseline classification algorithms. We carry out *9225* experiments in total.

Chongsheng Zhang and Paolo Soda contributed equally to this work.

✉ Gaojuan Fan
fangaojuan@126.com

Extended author information available on the last page of the article.

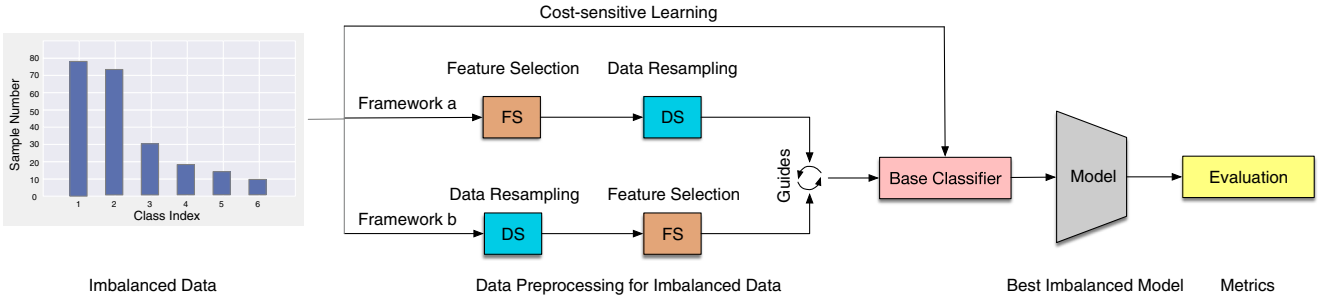


Fig. 1 The two frameworks we investigate for finding the best imbalanced classification model

Our proposal provides new findings and guidelines for deriving the best imbalanced classification model in practice (for applied data sciences). We reveal through extensive experiments that the synergy between feature selection and data resampling can yield significantly stronger models for imbalance learning. Moreover, we reveal that there is no constant winner between the two competing frameworks (i.e., FS+DS v.s. DS+FS); thus, both deserve attention when building imbalanced classification models. We also provide a detailed analysis on the influence of the baseline classifiers, the ratio between the number of majority and minority samples, and the ratio between the number of samples and features on the performance of imbalance classification.

The contributions of this work can be summarised as follows:

1. We reveal that it is worthwhile to try feature selection before and after data resampling to determine the optimal framework for imbalance classification; in particular, the feature selection after the data resampling framework should not be neglected as it was before.
2. We show that, under different evaluation metrics, the performance of these two frameworks varies for different baseline classifiers.
3. We provide recommendations of the Top-K specific combinations of feature selection and data resampling methods that are the top performers for every evaluation metric for each of the two frameworks.

The rest of this paper is organised as follows: we introduce the related work in Section 2, then formally define the problem to be investigated in this work in Section 3. We introduce the main methodology of this work in Section 4, present the experimental setup in Section 5, then report the results in Section 6. Finally, we conclude the paper in Section 7.

2 Related work

In this section, we review closely related works that investigate the joint influence of feature selection and data resampling approaches on imbalanced classification. For a general review on imbalance learning and feature selection, we refer the readers to survey papers such as [16].

Most approaches to multi-class imbalance classification are based on two-class imbalance classification algorithms. These approaches first decompose a multi-class dataset into several two-class datasets, next run two-class imbalance classification algorithms on each of them, then ensemble their outputs [33]. Furthermore, there is a shortage of benchmark datasets for multi-class imbalance classification, while many publicly available benchmark datasets exist for two-class imbalance learning. Due to the above reasons, we focus on two-class imbalance classification algorithms in this work.

In [14], the authors use 9 datasets to study the influence of feature selection and undersampling on software quality prediction. Their experimental results show that running feature selection on resampled data performs better than selecting the features from the original data. However, this work was developed for a very specific area; still, the authors only tested the RUS undersampling method [5] and a few filter feature selection methods.

In [15], the authors study and propose an iterative method for feature selection on imbalanced data. It first uses RUS undersampling technique to obtain balanced data. It next applies a filter-based feature selection technique to the resampled data and ranks all the features according to their predictive capacity. However, this study is limited to the use of RUS only and just one filter-based feature selection method. The authors admit that future work should consider more datasets and resampling techniques.

Wasikowski and Chen [29] examines the performance of various feature selection methods and how they combat the class imbalance problem. Besides using 7 filter feature selection methods (e.g. χ^2, information gain, Pearson correlation coefficient, signal-to-noise correlation coefficient), the authors also test the performance of one undersampling technique and one oversampling method (SMOTE) [7]. Their results show that the signal-to-noise correlation coefficient and the feature assessment by sliding thresholds are great candidates for feature selection on imbalanced data

with small sample size. However, the authors only test the feature selection before resampling strategy for class imbalance learning. Although they find that this strategy does not yield improved results, they admit this direction needs further investigation since they only test a few combinations of feature selection before resampling methods.

The authors in [27] study the impact of varying class imbalance ratios (IR for short) on the accuracy of a classifier. Empirical study in [18] shows that imbalance ratio is not the primary determinant in the class imbalance learning problem. A higher IR will further deteriorate the classification accuracy, but other data complexities also influence the classification performance.

It is worth noting that all the works mentioned above perform feature selection before resampling the data (FS+DS). None of them has explored the counterpart strategy, i.e. the feature selection after data selection framework (DS+FS). Furthermore, the experiments carried out in the existing works only use a very limited number of datasets, feature selection methods, and data resampling approaches. In view of the above limitations, this work extends our previous conference paper in [32] to provide a more comprehensive and in-depth empirical study and insight into the joint impact of feature selection and data resampling on imbalance classification.

Recent advances In order to adapt Synthetic Minority Oversampling Technique (SMOTE) to high-dimensional imbalanced classification settings, novel distance metrics for finding the neighbourhood of each minority sample were proposed in [19, 20]. In [25], the authors propose a decision tree ensemble model for imbalanced enterprise credit classification based on SMOTE and the Bagging ensemble learning algorithm with differentiated sampling rates.

In [23], the authors propose to concatenate two samples with the same labels into one sample for imbalanced learning, which transforms the original sample space into a concatenated new sample space.

3 Problem definition

This work aims to study how the performance of two-class imbalanced classification algorithms can be improved via the joint play of feature selection and data resampling methods. To this end, we compare the performance obtained by feature selection before the data resampling framework with the opposite (feature selection after data resampling) framework. These two frameworks are hereafter referred to as *FS+DS* and *DS+FS* for short.

The main research questions that we will investigate in this work are as follows:

1. **RQ1**: which framework, FS+DS or DS+FS, can achieve the best performance for two-class imbalance classification, and under which conditions?
2. **RQ2**: how does the classification performance vary when using DS+FS and FS+DS on imbalanced datasets with different imbalance ratios (IR) and samples-to-features ratios (SFR)?
3. **RQ3**: although there is no constant winner between the general FS+DS and DS+FS frameworks, for a given classifier, which specific combinations of feature selection and data resampling methods have a higher probability of becoming the top performers and thus be recommended with priority?

The first two issues are "macro-level" research questions, which investigate the overall comparative advantages of the two frameworks in two-class imbalance classification. The last issue is a "micro-level" research question, which aims to provide practical guidelines for choosing specific combinations of feature selection and data resampling methods for datasets of different characteristics.

4 Methodology

In this work, we aim to explore the joint use of data resampling and feature selection methods to create the best-performing imbalanced classification model. As shown in Fig. 1, we propose both the feature selection before the data resampling (*FS+DS*) framework and the feature selection after data resampling framework (*DS+FS*) to derive the best imbalanced classification model. In the following, we briefly introduce the specific data resampling and feature selection methods that will be used in these two frameworks.

4.1 Resampling approaches

Undersampling The undersampling methods considered in this work are Random Undersampling (RUS) [5], Condensed Nearest Neighbor decision rule (CNN) [12] and One-Sided Selection (OSS) [4]. Random Undersampling (*RUS*) balances class distribution through the random elimination of majority class examples until the desired class ratio between the minority and majority classes is reached [5]. Condensed Nearest Neighbor decision rule (*CNN*) successively discards instances (samples) that can be correctly classified using a prediction model built upon the current subset of instances [12]. One-sided Selection (*OSS*) detects and removes samples that are less reliable according to certain heuristics. The majority samples are divided into four groups, including noise, borderline, redundant, and safe samples, and only the safe majority samples are

combined with all the minority samples to create the final training set.

Oversampling The oversampling methods considered in this work are *SMOTE* [7], *SPIDER* [24], and *ADASYN* [13]. *SMOTE* stands for Synthetic Minority Oversampling Technique, which is an oversampling approach for generating synthetic samples in the feature space along the line segments joining pairs of minority class nearest neighbors. *SPIDER* is the acronym for Selective Preprocessing of Imbalanced Data. It oversamples misclassified instances of the minority class, while filtering difficult examples from the majority classes. *ADASYN* is short for ADAptive SYNthetic sampling. It is an improvement upon SMOTE, and the main idea is to use weighted distribution for different minority class examples according to their level of difficulty in learning.

4.2 Feature selection methods

In the literature, a large variety of feature selection approaches exist, which can be categorised into filter, wrapper, and embedded methods. In this work, we use 9 feature selection approaches presented below.

Filter methods Filter methods assess the relevance of features in the data. They usually compute a feature relevance score, removing features with low scores. Filter methods are independent of the classification algorithms and can easily scale to very high-dimensional datasets. Information gain (*Info*) evaluates the importance of an individual feature by measuring its corresponding information gain with respect to the class. χ^2 statistic (*ChiS*) evaluates the weight of an attribute by computing its chi-squared statistic value with respect to the class. Fisher feature selection (*Fish*) computes a score derived from the distances between data points in different classes, divided by the distances between data points of the same class. Gini Index (*Gini*) makes use of a statistical phenomenon called the Lorentz curve to quantify wealth/income inequality situations. Student's T-test (*T-test*) feature selection commonly uses Welch's T-test metric to measure a feature's capability in distinguishing two classes. Interested readers may refer to [17] for more detailed descriptions of these filter methods.

Wrapper methods Wrapper methods embed the model hypotheses within the feature subset search. The advantages of wrapper approaches lie in the interaction between feature subset search and model selection. The wrapper methods that we consider are Forward feature selection (*FWD*) [28], Linear forward selection (*Linear*) [10], and Ranked Search (*RS*) [11]. *FWD* feature selection first ranks each feature based on its relevance to the class label, then sequentially adds features to an empty candidate set until the addition of new features can not further improve the prediction performance [28]. Linear forward selection (*Linear*) improves the runtime FWD performance by limiting the attributes in the subsequent forward selection process to the top-ranked attributes [10]. In Rank Search (*RS*), subsets of increasing size (the best attribute plus the next best attribute) are evaluated, and the best attribute set is derived [11].

Embedded methods Embedded methods incorporate feature selection as part of the model building process. They enable interaction with the classification model, but one of the limitations is the computational load. In this work, we consider the Sparse Multinomial Logistic Regression with Bayesian regularisation method (*SBMLR*) [6], which realizes sparse feature selection by using a Laplace prior.

Time complexity discussion Most feature selection methods above have very high time complexity since they need to iteratively process the original data to derive the best feature subset. In contrast, data resampling methods such as SMOTE have comparatively low time complexity in general. Since the *DS+FS* framework first builds a small balanced subset of data and then performs feature selection on the subset, the computation time needed by feature selection methods can be remarkably reduced. Therefore, the *DS+FS* framework should have lower time complexity than the *FS+DS* framework.

5 Experimental setup

In this section, we introduce the datasets and evaluation metrics to be used in the experiments.

5.1 Datasets

In our experiments, we use 52 datasets belonging to real-world problems which are publicly available in the UCI and KEEL repositories [1, 2]. These datasets include *BioCells*, *cleveland*, *ecoli* variants, *german*, *glass* variants, *haberman*, *newthyroid*, *optdigit* variants, *pageblocks*, *pima*, *segment*, *semeion*, *winequality*, and *yeast* variants, etc.

We use *5*-fold stratified cross validation, considering the situation that resampling of the subsets introduces randomness. These *52* datasets, together with the use of *3* different classification algorithms, *9* feature selection methods, and *6* resampling methods, result in a large amount of experiments. Although a small proportion of experiments did not complete, primarily due to out-of-memory crashes or overly long running time, the final number of experiments is *9225*, which is still very large.

5.2 Baseline classification algorithms

We adopt three well-established classification algorithms that belong to different paradigms: C4.5 algorithm, a well-known decision tree based classification algorithm, Support-vector machine (SVM), a powerful yet flexible algorithm that is widely applied in supervised learning, and Multi-Layer Perceptron (MLP), an established artificial neural network for classification.

Although we acknowledge that the tuning of the hyper-parameters in the algorithms may lead to better results, *the No-Free-Lunch Theorems for Optimisation* tell us that all configurations perform equally well when averaged over all possible experiments [31]. Therefore, we adopt the default parameters so that the only way a method can outperform the other(s) relies on its fitness to the dataset. In our experiments, there is no hyperparameter to be specified in C4.5. For SVM, we use the LIBSVM implementation, the *precomputed* kernel type, and set the default regularisation parameter to *1.0*. For MLP, we fix the number of hidden layers to 10, with each layer having *100* units.

5.3 Evaluation metrics

A confusion matrix is commonly used to evaluate the performance of a classifier. Upon it, we use the following evaluation metrics to measure the performance of different methods under the two frameworks. Let N^+ and N^- be the number of minority (positive) and majority (negative) samples, respectively. TP and TN are the true positive and true negative results, while TPR and TNR are the corresponding rates. FP denotes the false positive results, Acc the accuracy, and p the precision.

$$Acc = \frac{TP + TN}{N^+ + N^-} \tag{1}$$

$$TPR = \frac{TP}{N^+} \tag{2}$$

$$TNR = \frac{TN}{N^-} \tag{3}$$

$$G\text{-}Mean = \sqrt{TPR \cdot TNR} \tag{4}$$

$$p = \frac{TP}{TP + FP} \tag{5}$$

$$\text{F1} = \frac{2}{1/p + 1/TPR} \tag{6}$$

$$IBA = (1 + \alpha \cdot (TPR - TNR)) \cdot TPR \cdot TNR \tag{7}$$

$G\text{-}Mean$, i.e., the geometric mean of per-class accuracies, is defined in (4). It increases when the accuracy of each class increases. As it is closely related to the distance to the perfect classification in the ROC space [3], we do not report the values of the area under the ROC curve.

The index of balanced accuracy (IBA) is a trade-off between the overall accuracy and the index of how balanced the two class accuracies are [9]. It intends to favour classifiers with better results on the minority class. IBA is computed using (7), with α set to 0.5 as suggested by the authors. Note that, if $\alpha = 0$, IBA turns into the square of $G\text{-}Mean$.

To measure the characteristics of a dataset, we also define the measures of imbalance ratio (IR) and samples-to-features ratio (SFR).

$$IR = \frac{N^-}{N^+} \tag{8}$$

$$SFR = \frac{N^+ + N^-}{m} \tag{9}$$

5.4 Statistical tests

Since we have a very large amount of experimental results, we need to use statistical analysis to find significant differences among different approaches or frameworks.

Pairwise comparisons We apply the Wilcoxon signed-rank test [22] which performs pairwise comparisons between the two frameworks and computes the differences between their performance scores. The two frameworks are ranked according to their absolute values in ascending order, and average ranks are assigned in cases of ties. Let R^+ be the sum of ranks for the datasets on which FS+DS outperforms DS+FS, and R^- be the sum of ranks for the opposite case (i.e., DS+FS outperforms FS+DS). If $min(R^+, R^-)$ is less than or equal to the value of the distribution of Wilcoxon for N_d degrees of freedom (N_d stands for the number of datasets), the null hypothesis of equality of means is rejected at the specified level of significance. We set $\alpha = 0.1$ not only based on the suggestions reported in [8], but also due to the large number of datasets and methods used in this work. To check whether the two classification approaches are significantly different and assess the degree of their difference, we also compute the p-value associated with each comparison, which represents the lowest level of significance of a hypothesis that results in a rejection.

To summarise the results, we also compute the quantity μ^+ and μ^-. The former denotes the number of times where R^+ outperforms R^- at the given significance level, while μ^- represents the opposite. These two values are further normalised with respect to the number of comparisons. This provides a synthetic index representing the overall predominance of an approach over the other.

Heuristic measures for Top-K specific combinations selections Given a specified dataset and evaluation metric, it is often appealing to have the recommendations on the Top-K specific combinations that are the most promising in yielding the best imbalanced classification performance, which can substantially reduce the workload in finding the best solution to imbalanced data classification. To this end, we provide two heuristic measures for recommending the Top-K specific combinations for imbalanced classification.

In the experiments, there are 52 datasets, and it is natural that some specific combinations (between the FS and DS methods) can only rank Top-K on some of the datasets. Our heuristic measure is to count the number of datasets (out of the 52 total datasets) that each specific combination can achieve the Top-K performance, in terms of a given evaluation metric (such as TPR). This heuristic measure is hereafter referred to as "Rank-SUM". The heuristic idea behind Rank-SUM is that the greater the Rank-SUM value for a specific combination, the more frequently (likely) this combination can become the Top-K performer on a dataset. In our experiments, we set K to 3 to limit the number of choices.

Besides, we also introduce the "Group-SUM" measure to check the overall performance of all the Top-K combinations (as a group). When using Group-SUM, we try each of the Top-K combinations in the group, then count the unique number of datasets where one of these Top-K combinations can become the Top-K method. The Top-K combinations can be complementary or redundant; if Group-SUM is close to the total sum of the Rank-SUM values of all the Top-K methods in the group, it implies that these Top-K methods are highly complementary, which is more desirable.

6 Results

6.1 Wilxcon tests for comparing the two frameworks

Figure 2 summarises via spider plots the number of times where the ranks of the Wilcoxon signed-rank test of the classifiers trained with FS+DS are larger than those obtained by DS+FS with the same classifier. In these charts, the blue and the orange lines correspond to FS+DS and DS+FS, respectively. The plot in the upper left of the figure shows the radar plot attained considering all the DS and FS methods. The second plot in the first row reports what happens when we combine only the oversampling approaches with all the feature selection methods. Similarly, the third plot in the same row reports what happens when we combine undersampling approaches with all the feature selection methods.

The second row of Fig. 2 shows the spider plots obtained when taking into account all the DS methods combined with different feature selection methods. Finally, the next six charts report the spider plots computed from the Cartesian product between the sets of DS and FS methods. For instance, the subfigure named as "only oversampling + only filters", which is shown as the first plot in the third row, summarises the number of times where the ranks of the Wilcoxon signed-rank test of a classifier, trained on data preprocessed by an oversampling method followed by a filter feature selection method (DS+FS), are larger than those obtained by the same classifier but with the FS+DS framework (in specific, the filter feature selection before oversampling method).

Visual inspection of Fig. 2 shows that FS+DS provides better results than DS+FS in most of the cases, irrespective of the performance metrics. In the case of $G\text{-}Mean$, $F1$ and IBA, preprocessing the training set with the FS+DS framework provides better performance than the DS+FS counterpart, since the blue lines in the spider plots are always external to the orange ones. But the opposite situation occurs in terms of accuracy in three plots, which is expected, since it is known that many resampling approaches improve the recognition performance on the minority class but negatively affect the accuracy of the majority to some extent.

Interestingly, the second and third plots in the first row also show that oversampling methods mitigate the decrease in accuracy with respect to undersampling, irrespective of the feature selection methods used. Furthermore, the third and fourth rows show that FS+DS outperforms DS+FS by a more significant extent when using oversampling rather than undersampling.

Another interesting discovery is that the ranking of filter methods such as Fisher is not affected by whether data resampling is applied or not; in contrast, wrapper and embedded methods are more sensitive to class-imbalance. Moreover, C4.5 is more sensitive to SVM and MLP in the class imbalance problem.

Summary Overall, there is no constant winner between the two frameworks. Thus, one should exhaustively test all the possible combinations of DS and FS, to derive the best imbalanced classification model for a specific data/application.

6.2 The impact of IR and SFR on imbalanced classification

We also investigate how the results vary with the imbalance ratio (IR) and the samples-to-features ratio (SFR). As mentioned before, IR is the ratio between the cardinality of the majority and the minority class, whereas SFR is the ratio between the number of samples and the number of

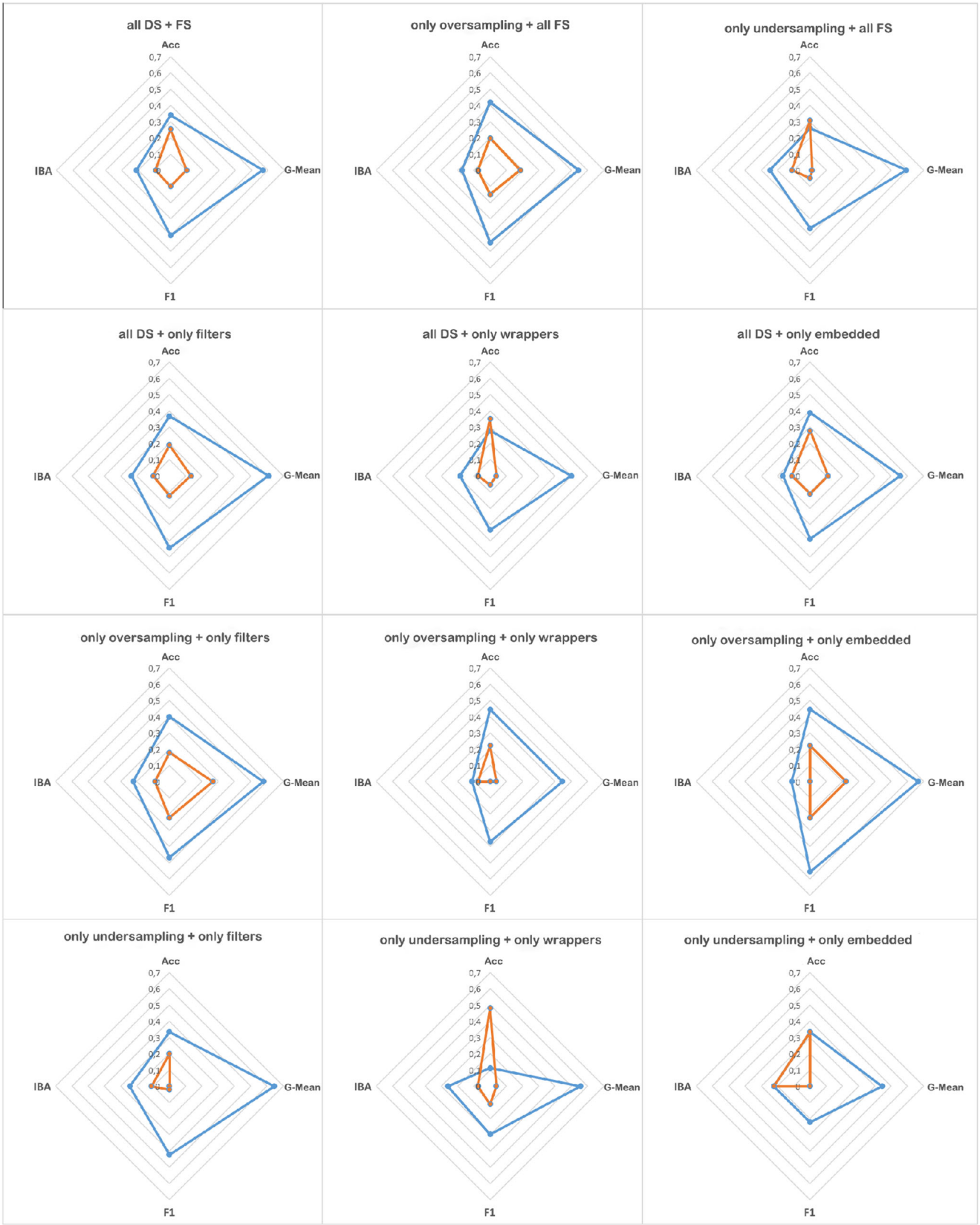


Fig. 2 Spider plot with values of μ^+ (in blue) and μ^- (in orange) for all the performance metrics; μ^+ and μ^- correspond to FS+DS and DS+FS, respectively

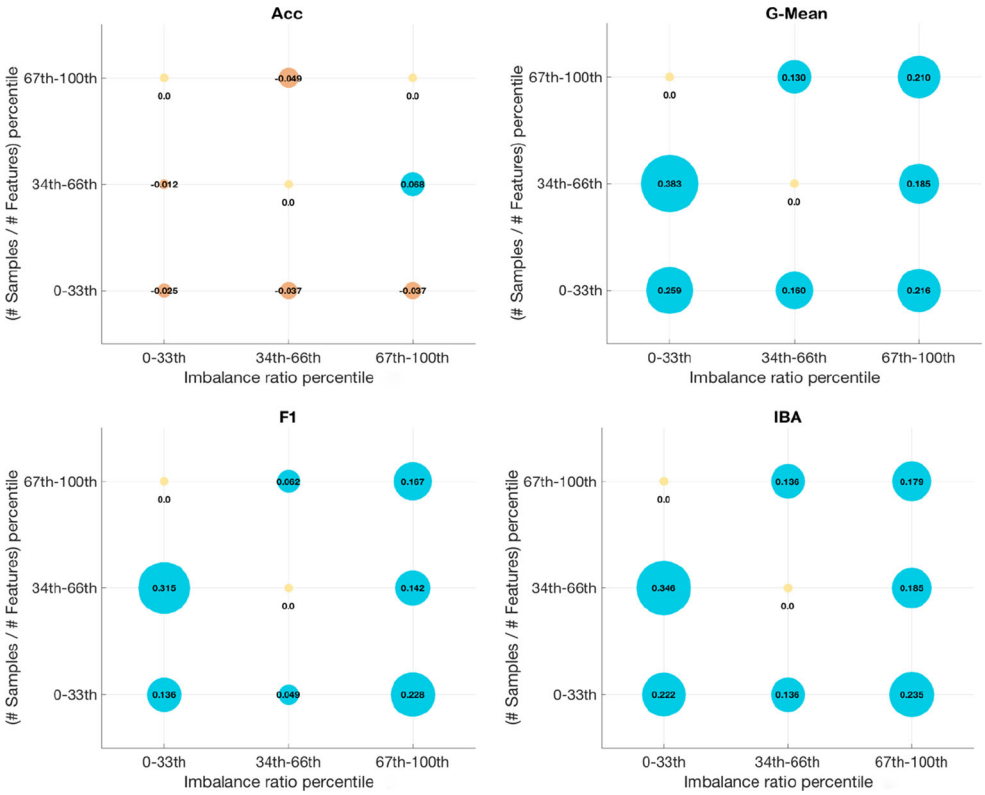


Fig. 3 Visual analysis of the predominance of FS+DS w.r.t. DS+FS for different pairs of IR and SFR, for each metric considered

features. For each of these two factors/ratios, we devide the experimental datasets into three groups: the first contains those falling below the 33th percentile, the second includes those between the 34th and the 66th percentile, and the third consists of those falling above the 66th percentile. The values of the 33th and the 66th percentile for IR are 6.01 and 9.18; but for SFR, the values are 37.3 and 115.09, indicating that SFR may have a significant influence on the performance of the two frameworks.

Figure 3 offers a visual presentation of the predominance of FS+DS over DS+FS for different pairs of IR and SFR, with different metrics considered. The predominance is measured in terms of μ^+ (μ^-), which calculates the number of times where R^+ outperforms (underperforms) R^- at the given significance level. On these grounds, given one of the three possible values of the percentile for both IR and SFR, the ball plots in the figure visualise the rate of predominance of FS+DS w.r.t. DS+FS. Positive values, coloured in blue, denote that the former outperforms the latter, whereas negative values, coloured in orange, have the opposite meaning. The results show that FS+DS outperforms DS+FS in general, in terms of the $G\text{-}Mean$, $F1$, and IBA. But in terms of Acc, we observe the predominance of DS+FS over FS+DS.

While these observations hold on datasets with high imbalance ratios, for datasets with large value of the SFR but small value of IR (see the upper left part of the plots), we notice that there is no predominance between the two frameworks. Such datasets are characterised by high SFRs and low IRs (the number of training samples is significantly larger than the number of features, while the ratio between the number of samples of majority and minority classes is not very large). Similar observations also hold for datasets with medium SFR and IR values, as seen from the central bubble in each plot of Fig. 3.

Summary For datasets with high SFR but relatively low IR ratios, the FS+DS and DS+FS frameworks return low, or zero values of μ^+ and μ^-, and it is hard to tell which framework is better. But for datasets with low SFR but

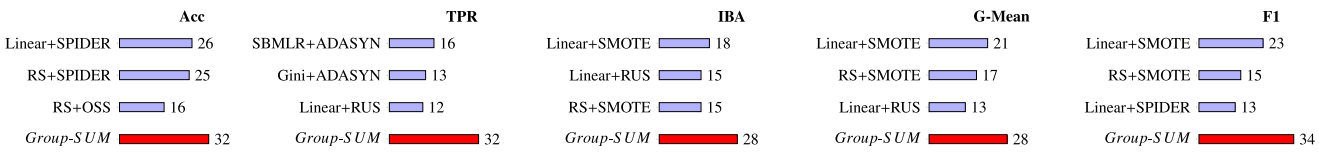


Fig. 4 Rank-SUM results for the Top-3 specific combinations under the FS+DS framework, using the C4.5 classifier

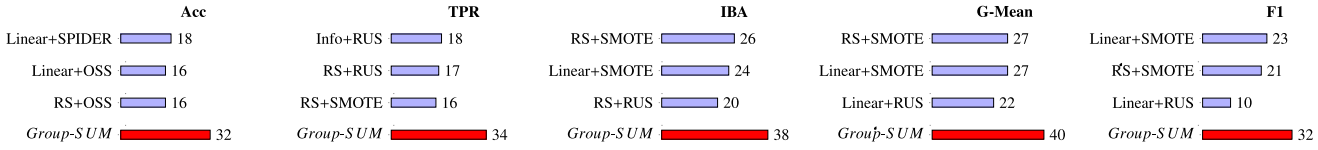


Fig. 5 Rank-SUM results for the Top-3 specific combinations under the DS+FS framework, using the C4.5 classifier

high IR ratios, the FS+DS framework has a comparative advantage over the DS+FS framework. However, once again, there is no constant winner between them.

6.3 Recommending the Top-K combinations with Rank-SUM

It is usually tempting for researchers and practitioners to have practical suggestions and guidelines about the specific top performing combinations for imbalanced classification on their specific data. In this context, we make a heuristic but intuitive analysis on the top-3 best-performing combinations between the FS and DS methods, using the heuristic measures we introduced for Top-K specific combination selection.

For each specific combination between the FS and DS methods (e.g., "Linear+SMOTE" with the C4.5 classifier), and under every different evaluation metric (e.g., $G\text{-}Mean$), we calculate its $Rank\text{-}SUM$ by counting the number of datasets where this combination ranks among the top-3 best performers. Then, for this metric, we select the group of the top-3 specific combinations in terms of $Rank\text{-}SUM$ (i.e. "Linear+SMOTE", "RS+SMOTE", and "Linear+RUS"), then calculate their $Group\text{-}SUM$. Note that, to facilitate the comparison of top performers under both frameworks, we still put the names of the feature selection methods before the data resampling methods in the DS+FS framework.

C4.5 Figure 4 presents the top-3 specific combinations between the DS and FS methods that achieve the highest "Rank-SUM" values, using the C4.5 classification algorithm and under the FS+DS framework. For the IBA, $G\text{-}Mean$ and F1 measures, "Linear+SMOTE" (in which the feature selection method is Linear, and the data resampling method is SMOTE) and "RS+SMOTE" are constantly among the top-3 combinations that yield the best $Rank\text{-}SUM$ values. Interestingly, this pattern still holds under the DS+FS framework using the same classifier, as depicted in Fig. 5. Therefore, these combinations can be recommended to the practitioners to try with priority. In addition, we see that Linear+SPIDER is constantly the best performer in terms of Acc, under both the FS+DS and DS+FS frameworks. But in terms of TPR, the top-3 specific combinations under the two frameworks are totally different. We also notice that the top-3 performers rank among the best performers only on half or less of the 52 datasets. Even if all the top-3 methods are tried, their $Group\text{-}SUM$ value indicates that, in union, they rank among the top-3 in less than 75% of the total datasets.

SVM Shown in Figs. 6 and 7, we see that, with SVM and under the $G\text{-}Mean$ and $F1$ metrics, the relative advantages of the top-3 combinations are not outstanding. Nevertheless, we can still find that RUS seems to be the best data resampling method in the two frameworks under all of the IBA, $G\text{-}Mean$ and TPR metrics. Overall, selecting the corresponding top performing combinations depends on the specific framework and evaluation metric.

Besides the Rank-SUM values, we also report the absolute cross-dataset averaged accuracy value for each of the top-3 combinations. That is, for each specific combination, we compute the averaged evaluation value of this method across all the 52 datasets. Shown in Figs. 8 and 9, we use box plots to demonstrate the cross-dataset average accuracy value of the top-3 performers when using the C4.5 classifier. This way, it is possible to intuitively check the general performance of each specific combination under different evaluation metrics. Similarly, in Figs. 10 and 11, we also present the cross-dataset average accuracy values for SVM. It is clear that, in terms of the $G\text{-}Mean$ metric and with SVM, the top-3 performers under the DS+FS framework slightly outperform their counterparts under the FS+DS framework.

With the absolute cross-dataset average evaluation values, it is also possible to roughly compare the overall performance of different combinations across different classifiers. For instance, as shown in Figs. 9 and 11, in terms of IBA and $G\text{-}Mean$, the top-3 combinations under C4.5

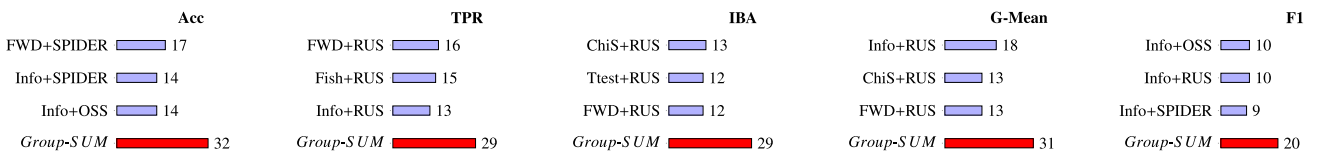


Fig. 6 Rank-SUM results for the Top-3 specific combinations under the FS+DS framework, using the SVM classifier

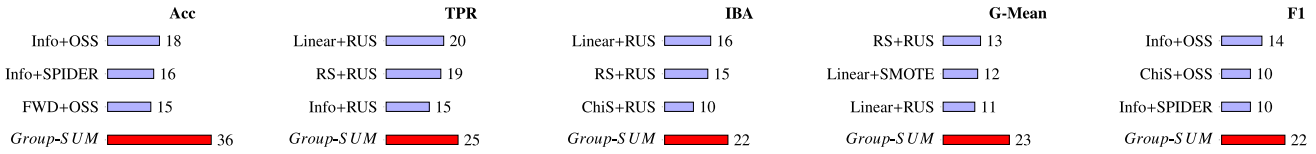


Fig. 7 Rank-SUM results for the Top-3 specific combinations under the DS+FS framework, using the SVM classifier

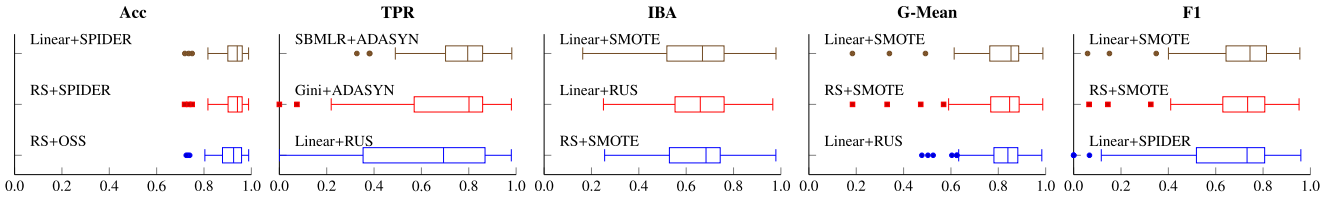


Fig. 8 Box plots for the cross-datasets averaged value of each metric, for the Top-3 specific combinations using C4.5 and FS+DS framework

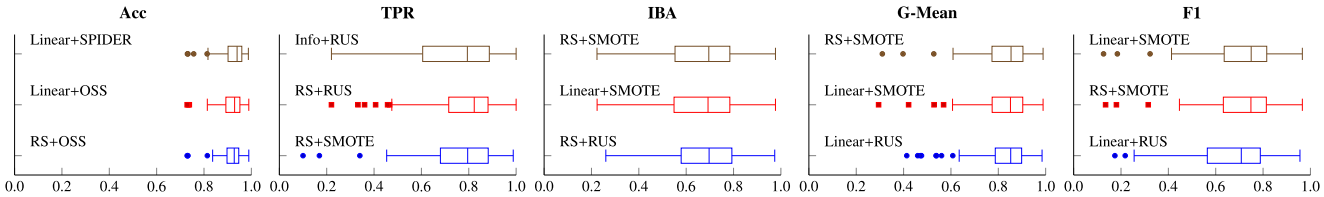


Fig. 9 Box plots for the cross-datasets averaged value of each metric, for the Top-3 specific combinations using C4.5 and DS+FS framework

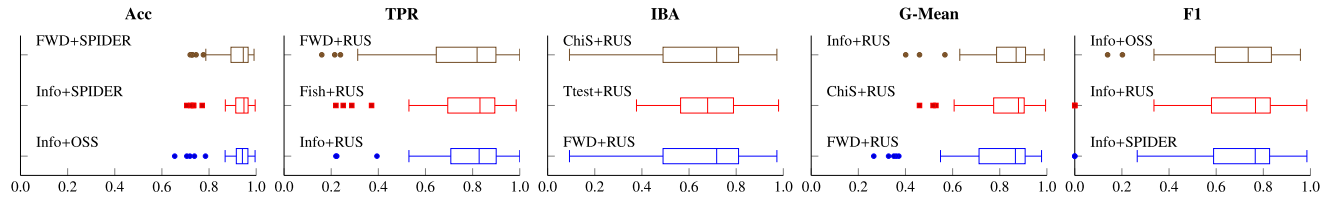


Fig. 10 Box plots for the cross-datasets averaged value of each metric, for the Top-3 specific combinations using SVM and FS+DS framework

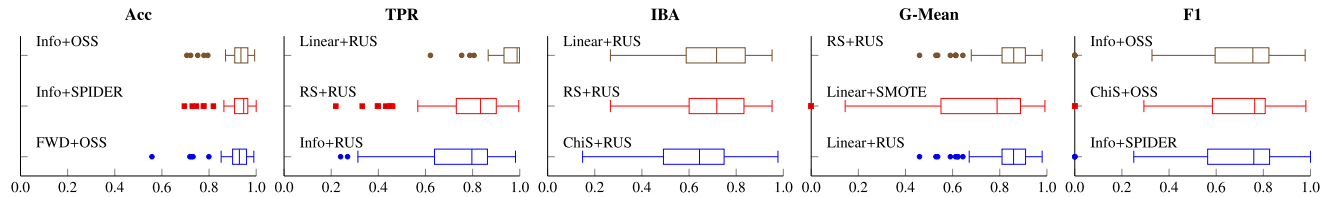


Fig. 11 Box plots for the cross-datasets averaged value of each metric, for the Top-3 specific combinations using SVM and DS+FS framework

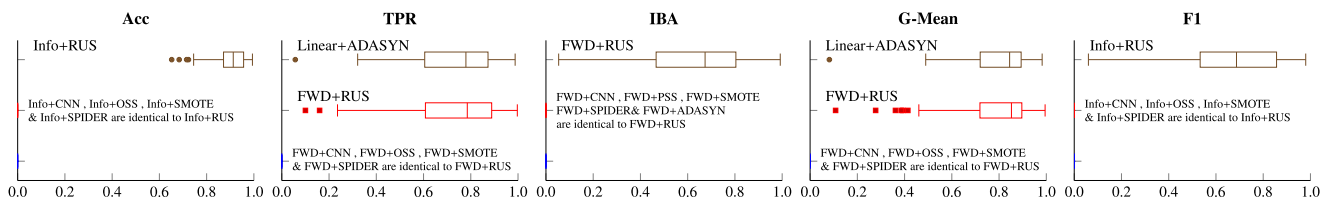


Fig. 12 Box plots for the cross-datasets averaged value of each metric, for the Top-3 specific combinations using MLP and FS+DS framework

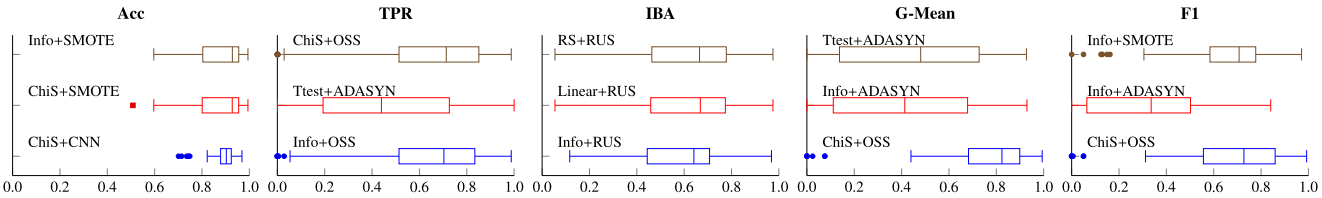


Fig. 13 Box plots for the cross-datasets averaged value of each metric, for the Top-3 specific combinations using MLP and DS+FS framework

and the DS+FS framework, and the top-3 combinations under SVM and the same framework, yield remarkably better performance than the others.

MLP For MLP under the FS+DS framework, the patterns are quite different from C4.5 and SVM. From Fig. 12, we observe that, in terms of *G-Mean*, the best performer is Linear+ADASYN, while in terms of IBA, the best method is FWD+RUS. But for the other combinations, the influence of data resampling on the performance is insignificant, and the feature selection method matters most. For instance, in terms of IBA, the best feature selection method is constantly FWD; in terms of *G-Mean*, except the top-1 combination (Linear+ADASYN), FWD is also the best feature selection method for the rest top-performing methods/combinations. Still, in terms of F1, Info is constantly the best feature selection method for all the top performers. Therefore, for MLP under the FS+DS framework, we can resort to a two-stage method for efficiently deriving the best combination. In the first stage, we can find which feature selection methods yield the best performance for a given evaluation metric. In the second stage, we only need to combine the selected feature selection method(s) with different resampling methods. This way, the computation cost can be substantially reduced. However, the above patterns do not hold for MLP with the DS+FS framework. From Fig. 13, we find that Chis+OSS is generally the winner in terms of *G-Mean*, TPR, and F1, since it obtains the best cross-dataset averaged value for each of these three metrics. But for IBA under the same DS+FS framework, the best combination is RS+RUS, and RUS is the best data resampling method for all the top-3 combinations (Fig. 13).

Summary We find that for IBA under the C4.5 classifier, the specific combinations of SMOTE+RS and SMOTE+Linear under the DS+FS framework, and Linear+RUS and Linear+RUS under the FS+DS framework, are among the top-3 best performers. For SVM, the specific combinations of ChiS+RUS and Ttest+RUS under the FS+DS framework are among the top-3 performers for both measures. For MLP, RUS+RS and RUS+Linear under the DS+FS framework, and FWD+RUS and FWD+CNN under the FS+DS framework, are among the top performers.

7 Conclusions

This work investigates the feature selection before or after data resampling frameworks for improving two-class imbalanced classification. We discover that, in order to derive the best performing imbalanced classification model, both frameworks should be considered. Moreover, we reveal that the choice of the two frameworks depends on the specific dataset, the combinations of feature selection and data resampling algorithms, as well as the baseline classification algorithms. Finally, we also suggest the Top-K specific combinations that often obtain the best imbalanced classification performance.

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Affiliations

Chongsheng Zhang[1] · Paolo Soda[2,3] · Jingjun Bi[1] · Gaojuan Fan[1] · George Almpanidis[1] · Salvador García[4] · Weiping Ding[5]

Chongsheng Zhang
cszhang@ieee.org

Paolo Soda
p.soda@unicampus.it

Jingjun Bi
bi.jingjun@outlook.com

George Almpanidis
almpanidis@gmail.com

Salvador García
salvagl@decsai.ugr.es

Weiping Ding
dwp9988@163.com

1 Henan Key Lab of Big Data Analysis and Processing, Henan University, Henan, China

2 Department of Engineering, University Campus Bio-Medico of Rome, Rome, Italy

3 Department of Radiation Sciences, Radiation Physics, Biomedical Engineering, Umea University, Umea, Sweden

4 DaSCI Andalusian Research Institute, University of Granada, Granada, Spain

5 School of Information Science and Technology, Nantong University, Nantong, China